



```
In [4]: import numpy as np
import pandas as pd

import tensorflow as tf

from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.applications import MobileNetV2
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D, Dropout
from tensorflow.keras.models import Model
from tensorflow.keras.optimizers import Adam
```

```
In [13]: from tensorflow import keras
max_features = 10000
maxlen = 200

(X_train, y_train), (X_test, y_test) = keras.datasets.imdb.load_data(num_words=10000)

X_train = keras.preprocessing.sequence.pad_sequences(X_train, maxlen=maxlen)
x_test = keras.preprocessing.sequence.pad_sequences(x_test, maxlen=maxlen)
```

```
In [14]: from tensorflow.keras import layers
def make_rnn():
    return keras.Sequential([
        layers.Embedding(max_features, 128, input_length=maxlen),
        layers.SimpleRNN(128),
        layers.Dense(1, activation='sigmoid')
    ])

def make_lstm():
    return keras.Sequential([
        layers.Embedding(max_features, 128, input_length=maxlen),
        layers.LSTM(128),
        layers.Dense(1, activation='sigmoid')
    ])

def make_gru():
    return keras.Sequential([
        layers.Embedding(max_features, 128, input_length=maxlen),
        layers.GRU(128),
        layers.Dense(1, activation='sigmoid')
    ])

models = {
    "RNN": make_rnn(),
    "LSTM": make_lstm(),
    "GRU": make_gru()
}
```

```
In [15]: for name, model in models.items():
    model.compile(
        optimizer='adam',
        loss='binary_crossentropy',
        metrics=['accuracy'])
```

```
)  
print(f"{name} compiled!")
```

RNN compiled!
LSTM compiled!
GRU compiled!

```
In [16]: print(X_train.dtype)  
         print(type(X_train))
```

int32
<class 'numpy.ndarray'>

```
In [17]: history = {}  
  
         for name, model in models.items():  
             print(f"\nTraining {name}...\n")  
  
             history[name] = model.fit(  
                 X_train, y_train,  
                 epochs=5,  
                 batch_size=64,  
                 validation_split=0.2,  
                 verbose=1  
             )
```

Training RNN...

Epoch 1/5

```
2026-04-08 14:50:52.753630: I external/local_xla/xla/service/service.cc:163] XLA service 0x7d454001ac40 initialized for platform CUDA (this does not guarantee that XLA will be used). Devices:
2026-04-08 14:50:52.753660: I external/local_xla/xla/service/service.cc:171] StreamExecutor device (0): NVIDIA GeForce RTX 3060 Laptop GPU, Compute Capability 8.6
2026-04-08 14:50:52.824949: I tensorflow/compiler/mlir/tensorflow/utils/dump_mlir_util.cc:269] disabling MLIR crash reproducer, set env var `MLIR_CRASH_REPRODUCER_DIRECTORY` to enable.
2026-04-08 14:50:53.090266: I external/local_xla/xla/stream_executor/cuda/cuda_dnn.cc:473] Loaded cuDNN version 91002
2026-04-08 14:50:53.142086: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.
2026-04-08 14:50:53.142132: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.
2026-04-08 14:50:53.142150: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.
2026-04-08 14:50:53.142163: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.
2026-04-08 14:50:56.890732: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function 'gemm_fusion_dot_34', 292 bytes spill stores, 292 bytes spill loads
```

2/313 ————— **28s** 91ms/step - accuracy: 0.4844 - loss: 0.6912

I0000 00:00:1775638258.182347 1782 device_compiler.h:196] Compiled cluster using XLA! This line is logged at most once for the lifetime of the process.

311/313 ————— **0s** 55ms/step - accuracy: 0.5682 - loss: 0.6667

2026-04-08 14:51:15.592479: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.

2026-04-08 14:51:15.592532: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.

2026-04-08 14:51:15.592548: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.

2026-04-08 14:51:15.592560: I external/local_xla/xla/service/gpu/autotuning/dot_search_space.cc:208] All configs were filtered out because none of them sufficiently match the hints. Maybe the hints set does not contain a good representative set of valid configs? Working around this by using the full hints set instead.

2026-04-08 14:51:17.477867: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function 'gemm_fusion_dot_32', 36 bytes spill stores, 36 bytes spill loads

2026-04-08 14:51:18.934172: I external/local_xla/xla/stream_executor/cuda/subprocess_compilation.cc:346] ptxas warning : Registers are spilled to local memory in function 'gemm_fusion_dot_31', 36 bytes spill stores, 36 bytes spill loads

313/313  **31s** 78ms/step - accuracy: 0.6196 - loss: 0.6328 - val_accuracy: 0.7804 - val_loss: 0.4919
 Epoch 2/5
313/313  **21s** 66ms/step - accuracy: 0.7833 - loss: 0.4623 - val_accuracy: 0.7922 - val_loss: 0.4877
 Epoch 3/5
313/313  **15s** 48ms/step - accuracy: 0.8329 - loss: 0.3832 - val_accuracy: 0.7090 - val_loss: 0.5601
 Epoch 4/5
313/313  **13s** 42ms/step - accuracy: 0.8699 - loss: 0.3189 - val_accuracy: 0.7960 - val_loss: 0.4965
 Epoch 5/5
313/313  **15s** 46ms/step - accuracy: 0.8974 - loss: 0.2589 - val_accuracy: 0.7630 - val_loss: 0.5373

Training LSTM...

Epoch 1/5
313/313  **12s** 31ms/step - accuracy: 0.7611 - loss: 0.4839 - val_accuracy: 0.8538 - val_loss: 0.3499
 Epoch 2/5
313/313  **11s** 36ms/step - accuracy: 0.8899 - loss: 0.2735 - val_accuracy: 0.8598 - val_loss: 0.3202
 Epoch 3/5
313/313  **8s** 27ms/step - accuracy: 0.9186 - loss: 0.2159 - val_accuracy: 0.8472 - val_loss: 0.4823
 Epoch 4/5
313/313  **8s** 25ms/step - accuracy: 0.9366 - loss: 0.1702 - val_accuracy: 0.8710 - val_loss: 0.3727
 Epoch 5/5
313/313  **10s** 33ms/step - accuracy: 0.9577 - loss: 0.1186 - val_accuracy: 0.8596 - val_loss: 0.4442

Training GRU...

Epoch 1/5
313/313  **10s** 30ms/step - accuracy: 0.7542 - loss: 0.4882 - val_accuracy: 0.8514 - val_loss: 0.3505
 Epoch 2/5
313/313  **8s** 24ms/step - accuracy: 0.8815 - loss: 0.2942 - val_accuracy: 0.8676 - val_loss: 0.3276
 Epoch 3/5
313/313  **9s** 30ms/step - accuracy: 0.9226 - loss: 0.2006 - val_accuracy: 0.8664 - val_loss: 0.3400
 Epoch 4/5
313/313  **10s** 32ms/step - accuracy: 0.9531 - loss: 0.1309 - val_accuracy: 0.8602 - val_loss: 0.3952
 Epoch 5/5
313/313  **8s** 26ms/step - accuracy: 0.9710 - loss: 0.0832 - val_accuracy: 0.8680 - val_loss: 0.4486

```
In [18]: results = {}

         for name, model in models.items():
```

```
print(f"\nEvaluating {name}...\n")

loss, acc = model.evaluate(x_test, y_test, verbose=0)
results[name] = acc

print(f"{name} Test Accuracy: {acc:.4f}")
```

Evaluating RNN...

RNN Test Accuracy: 0.7588

Evaluating LSTM...

LSTM Test Accuracy: 0.8535

Evaluating GRU...

GRU Test Accuracy: 0.8628

In []: